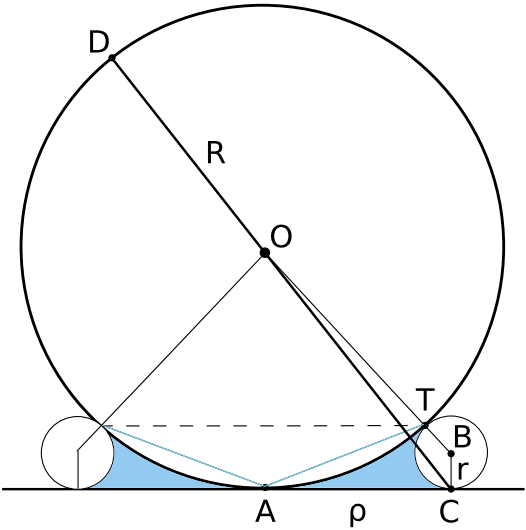
3. Sticky ball (4 points) — *Solution by Jaan Kalda, Tudor Plopeanu, grading schemes by Eppu Leinonen.*



Let the radius of the neck be $\rho \ll R$ thus the force from surface tension at the contact of the sphere is negligible (angle to the horizontal $\approx$ 0). Then, the curvature radius of the meniscus $r \ll \rho$ can be found from the intersecting secants theorem applied on a point C on the bottom edge of the neck relative to the ball and approximating that the point T is (near) the intersection of CD and the circle: $2r \cdot 2R \approx \rho^2$. On the drawing, $AC^2 \approx CT \cdot CD$ and we approximate CT as $2r$, $R + r$ as R, and AC as ρ. The pressure difference between the water and the surrounding air is $\Delta p = \sigma/r$. Its total vertical component is equal to its amplitude times the vertical cross-section $S = \pi\rho^2$. As such, the total vertical force on the ball difference between the two cases (with and without water) is $\Delta F \approx \sigma\pi\rho^2/r = 4\pi\sigma R$.

Grading: (preliminary) Note that differing sign conventions are tolerated

- Stating that the meniscus is (roughly) inverse spherical or usage of constant r to characterise the meniscus as inverse spherical **(0.5 pts)**
- $r \ll \rho \ll R$ **(0.1 pts)**
- Stating that surface tension from from contact is negligible or ΔF comes from the pressure difference (either explicitly or implicitly) **(0.5 pts)**
- $2r \cdot 2R \approx \rho^2$ **(0.7 pts)** (if not found, partial points can be earned as below)

  - $CT \approx 2r$ **(0.1 pts)**
  - $R + r \approx R$ **(0.1 pts)**
  - $AC \approx \rho$ **(0.2 pts)**
  - $AC^2 \approx CT \cdot CD$ or statement of intersecting secants theorems **(0.3 pts)**
- $\Delta p = \sigma/r$ **(1 pts)** (if not found, partial points can be earned as below)
  - $\Delta p = \sigma(1/r_1 + 1/r_2)$ or any attempt to use a form of the Laplace-Young equation **(0.2 pts)**
  - $\Delta p = \sigma(1/\rho - 1/r)$ **(0.3 pts)**
- $S = \pi\rho^2$ (correct effective are for pressure) **(0.5 pts)**
- $\Delta F = S\Delta p$ **(0.5 pts)**
- $\Delta F \approx 4\pi\sigma R$ **(0.2 pts)** (0.1 pts for each of the following)
  - $\Delta F > 0$ or noting that the contact force increases
  - Correct dimensionless factor of 4π and correct dimensions (only if approach is correct)

Solution 2 by Eppu Leinonen

The position of the meniscus can be parameterised using the angle $\theta = \angle TOA$. The meniscus is small as the amount of water is small. Thus $|\theta| \ll 1$ and the radii can be approximated as $\rho = R\sin\theta \approx R\theta$ and $2r \approx R(1 - \cos\theta) \approx R\theta^2/2$. The change in normal force due to surface tension is $\Delta F_\sigma = 2\pi\sigma\rho\sin\theta \approx 2\pi\sigma R\theta^2 \approx 0$. Using the Young-Laplace equation $\Delta p = \sigma(1/\rho - 1/r) \approx \sigma(1/\theta - 4/\theta^2)$. Thus the total change in normal force due to pressure difference is $\Delta F_p = -S\Delta P = -\pi\rho^2\Delta P \approx \pi\sigma R(4 - \theta) \approx 4\pi\sigma R^2$. I.e. $\Delta F = \Delta F_p + \Delta F_\sigma \approx \Delta F_p \approx 4\pi\sigma R$.

Grading: (preliminary) Note that differing sign conventions are tolerated

- Stating that the meniscus is (roughly) inverse spherical or usage of constant r to characterise the meniscus as inverse spherical **(0.5 pts)**
- $|\theta| \ll 1$ **(0.1 pts)**
- $\rho \approx R\theta$ **(0.2 pts)**
- $AC \approx \rho$ **(0.2 pts)**
- $r \approx R\theta^2/4$ **(0.3 pts)**
- $\Delta F_\sigma \propto \theta^2 \ll \Delta F_p$ or any statement that the force from surface tension is negligible or $\Delta F = \Delta F_p$ **(0.5 pts)**

- $\Delta p = \sigma/r$ **(1 pts)** (if not found, partial points can be earned as below)
  - $\Delta p = \sigma(1/r_1 + 1/r_2)$ or any attempt to use a form of the Laplace-Young equation **(0.2 pts)**
  - $\Delta p = \sigma(1/\rho - 1/r)$ **(0.3 pts)**
- $S = \pi\rho^2$ (correct effective area for pressure) **(0.5 pts)**
- $\Delta F_p = -S\Delta p$ **(0.5 pts)**
- $\Delta F \approx 4\pi\sigma R$ **(0.2 pts)** (0.1 pts for each of the following)
  - $\Delta F > 0$ or noting that the contact force increases
  - Correct dimensionless factor of 4π and correct dimensions (only if approach is correct)

Note: A common approach was to use the method of virtual displacement to solve for the force caused by the meniscus. However, it turns out that $\delta A_{lg} \propto \sqrt{r}\delta r$ (the area of the meniscus-air interface is roughly half of a spherical toroid with radii ρ and r) which means that the change in potential energy due to the change in surface area is negligible ($r \ll R$). Thus the virtual work done by lifting the sphere only counteracts the virtual work done by the pressure, which reduces the problem back to either solution 1 or solution 2. Thus no points are granted for just mentioning this approach and telling that $\delta U = \sigma\delta A_{lg}$.

Solution 3 using virtual displacement (by Jaan Kalda). Let us denote the contact area of water and plate with $A = \pi\rho^2$; then, the contact area of water and ball is also approximately A. Let the ball touch initially the plate, and then be raised by $\mathrm{d}x$. Since the volume of water is conserved, no work is made by atmospheric pressure, and $2r\mathrm{d}A = A\mathrm{d}x$, where $\mathrm{d}A$ is the change of the contact area; hence, $\frac{\mathrm{d}A}{\mathrm{d}x} = A/2r$. Meanwhile, for zero contact angle, the difference of surface energies at the air-solid interface, and at the water-solid interface equals to the surface energy of the water-air interface. This fact can be expressed in terms of the three surface tension coefficients denoted with σ_1, σ_2, and σ, respectively: $\sigma_1 - \sigma_2 = \sigma\cos\alpha = \sigma$. During our virtual displacement, the air-water interface remains almost constant; meanwhile,

air-solid interface is increased by $2\mathrm{d}A$ (contributed equally by the ball and plate surfaces). Therefore, the surface energy is increased by $\mathrm{d}U = (\sigma_1 - \sigma_2)2\mathrm{d}A = 2\sigma\mathrm{d}A$. Now we can find force as $F = \frac{\mathrm{d}U}{\mathrm{d}x} = 2\sigma\frac{\mathrm{d}A}{\mathrm{d}x} = \sigma A/r$. This is the same expression we obtained from Young-Laplace equation. From this point on, the solution follows the steps made above.

Grading: (preliminary) Note that differing sign conventions are tolerated.

- Only the contact area between the solid and the liquid changes notably **(0.5 pts)**
- $\frac{\mathrm{d}A}{\mathrm{d}x} = A/2r$ **(0.5 pts)**
- $\sigma_1 - \sigma_2 = \sigma$ **(0.2 pts)**
- $\mathrm{d}U = 2(\sigma_1 - \sigma_2)\mathrm{d}A$ **(0.3 pts)**
- $F = \frac{\mathrm{d}U}{\mathrm{d}x}$ (only if A has been identified correctly) **(0.3 pts)**
- $F = \sigma A/r$ **(0.2 pts)**

These replace noting that $\Delta F_\sigma \approx 0$, $\Delta p = \sigma/r$ and $\Delta F = S\Delta p$ from the other solutions.

4. Totality (8 points) — *Solution by Tudor Plopeanu, Taavet Kalda.* **Grading:** (preliminary)

- In in this task, numerical errors in the answers give **(-0.1 pts)** (as long as the answer is reasonable)

i) *(1.5 points)* As the distance from the Moon to the Sun is much larger than the distance from the Moon to the Earth, we can consider the speed of the Moon's shadow on Earth to be equal to the Moon's speed altogether. The speed of the Moon is the ratio of the circumference of its orbit to its period:

$$v_m = \frac{2\pi R_m}{T_m} = 1.02\,\mathrm{km/s}.$$

Because during the peak of the eclipse, the centre-points of Earth, the Moon and the Sun lie on the same line, the shadow of the Moon must move along the diameter of the Earth when viewed face-on. As such, the shadow travels a distance of $2r_e$ while it's still visible on Earth and the eclipse will be observable

for time

$$T_{\text{ecl}} = \frac{2r_e}{v_m} = \frac{r_e}{\pi R_m} T_m \approx 12.5 \times 10^3\,\text{s} = 3.46\,\text{h}.$$

Grading: (preliminary)

- Explains that the moons shadow can be approximated by the moons position **(0.5 pts)**
- Correct speed of the moon **(0.5 pts)**
- Correct final expression **(0.5 pts)**
- Minor mistake in final expression **(-0.2 pts)**

ii) *(1 point)* In absence of the Earth's rotation, the Moon's shadow would cover close to π radians. Over T_{ecl}, the Earth rotates by $2\pi\frac{T_{ecl}}{T_0}$ radians. The Earth's rotation goes "in the same direction" as the Moon's shadow, and therefore the total longitudinal reach of the eclipse is $|180(1 - 2T_{\text{ecl}}/T_0)| \approx 128$ degrees.

Grading: (preliminary)

- Figures out that the angle is π without the rotation of the earth **(0.3 pts)**
- Explains that the angle becomes smaller than π due to earths rotation **(0.3 pts)**
- Correct final formula and answer **(0.4 pts)**
- Thinks earths rotation goes against the moons rotation and obtains an angle $> \pi$ **(-0.3 pts)**

iii) *(1.5 points)* Right near the end of the eclipse, the Earth's rotation moves the surface under the Moon's shadow almost perpendicularly to the Moon's shadow's velocity. As such, the width of the Moon's shadow is equal to the distance travelled by the Moon's shadow for the duration of the totality at that point,

$$w_\lambda = v_m t_0 = 2\pi R_m \frac{t_0}{T_m} \approx 123\,\text{km}.$$

At the equator, the surface of the Earth is closer by r_e to the Moon compared to the point where the eclipse ends, serving to further increase the area of the full shadow. As the Moon covers an angle of approximately $\alpha = 2r_m/R_m$ radians in the sky, when getting closer to the moon by r_e, the width of its shadow will increase by roughly $\alpha r_e \approx$

57.7 km. As such, the total width of the shadow near the equator is around $w_{\text{eq}} = v_m t_0 + \alpha r_e = 180\,\text{km}$.

Grading: (preliminary)

- Realizes that the velocities are perpendicular at the end of the eclipse **(0.4 pts)**
- Calculates the width of the eclipse at some point on earth **(0.5 pts)**
- Correctly finds how to translate the width to the width at the equator **(0.5 pts)**
- Correct final answer **(0.1 pts)**

iv) *(1.5 points)* The peak of the eclipse occurs on the equator, firstly because that's where the Moon's shadow is the biggest due to it being closer to the Moon, but also because the Earth's rotation is the fastest there, and the rotation of the Earth serves to lengthen the effect of the eclipse. At the equator the surface of the Earth moves with speed $v_e = 2\pi r_e/T_0 = 0.46\,\text{km/s}$. The angle between the velocity of the Earth's surface at the equator and the Moon's shadow's velocity is λ, so the relative speed of the Moon's shadow can be obtained from cosine theorem as

$$v_{\text{rel}} = \sqrt{v_m^2 + v_e^2 - 2v_m v_e \cos\lambda} = 0.654\,\text{km/s}.$$

The duration while the eclipse can be observed during the peak is thus $t_{\text{eq}} = w_{\text{eq}}/v_{\text{rel}} = 276\,\text{s} = 4.6\,\text{min}$.

Grading: (preliminary)

- Shows that the eclipse must be observable the longest time at the equator **(0.5 pts)**
- Finds the relative velocity between the moon's shadow and the velocity of the surface of the earth **(0.5 pts)**
- From this finds the formula for the maximum duration of the eclipse **(0.5 pts)**
- Forgetting to take the angle between the velocities into account **(-0.3 pts)**

v) *(1 point)* Because $a \ll R_e$, we can treat the relative velocity between the Moon's shadow and Earth to be the same as in the previous subpart. The full shadow of the Moon on the Earth at the equator is a circle, and we can find the distance the Moon's shadow has to travel over the course of the totality at a point displaced by a using the Pythagorean Theorem as $2(w_{\text{eq}}^2/4 - a^2)^{1/2}$. The eclipse is then

observable for

$$\frac{1}{v_{\text{rel}}}\sqrt{w_{\text{eq}}^2 - 4a^2} = 230\,\text{s} = 3.8\,\text{min}.$$

Grading: (preliminary)

- Argues that the relative velocity is approximately the same as in the previous task **(0.5 pts)**
- Correct geometry and answer **(0.5 pts)**

vi) *(1.5 points)* With our assumptions, the typical width of an eclipse is $(w_\lambda + w_{\text{eq}})/4$, and the typical length on the surface of the Earth is πr_e. As such, a typical eclipse covers an area equal to the product of the two above values. From there, we deduce the probability of an eclipse being noticeable from a given arbitrary point on Earth as the area spanned by the eclipse, $\pi r_e(w_\lambda + w_{eq})/4$, divided by the surface area of Earth, $4\pi r_e^2$. Then, the expected number of eclipses until one covers a given arbitrary point is the multiplicative inverse of the above probability. As eclipses occur, on average, every 18 months, the expected time for this to happen is

$$\frac{16 r_e}{w_\lambda + w_{\text{eq}}} \cdot 18\,\text{months} \approx 500\,\text{years}.$$

Grading: (preliminary)

- Shows that the probability per eclipse is equal to the area covered by the eclipse divided by the total area of the earth **(0.5 pts)**
- Finds the area covered by one eclipse **(0.5 pts)**
- Correctly multiples the inverse probability with the duration between eclipses and finds the correct answer **(0.5 pts)**

5. STRING AND PENDULUM (10 points) — *Solution by Päivo Simson, Tudor Plopeanu.*

i) *(5 points)* We build two pendulums of different lengths and release them from different angles such that their periods are equal. We choose a small enough amplitude for the longer pendulum to be able to use the small angle formula, and a large amplitude for the short pendulum.

We measure the lengths of the pendulums. Let l_0 be the length of the longer pen-

dulum and l_1 be the length of the shorter pendulum. Additionally, we measure the angular amplitude ϕ (in degrees) of the shorter (larger-amplitude oscillating) pendulum with a protractor. From the equality of periods, we get

$$\sqrt{l_0} = \sqrt{l_1}(1 + A\alpha^2),$$

from which

$$A = \frac{\sqrt{\frac{l_0}{l_1}} - 1}{\alpha^2} = \frac{\sqrt{\frac{l_0}{l_1}} - 1}{\left(\frac{\pi\phi}{180^\circ}\right)^2}.$$

We repeat the experiment with different lengths and finally find the average of the results. With the example values: $\phi = 55^\circ$, $l_0 = 41.5$ cm and $l_1 = 36.5$ cm we get $A \approx 0.07$. The theoretical true value is $A \approx 0.063$, but the expected measured value is slightly larger, as higher order terms of the theoretical expansion $(1 + A\alpha^2 + B\alpha^4 + \dots)$ are "combined" in the value of A. Thus generally, the larger the angle used in the measurement, the larger the value of A, which is why a value in the range $A \in [0.06, 0.08]$ is to be expected.

We note a common solution attempt of measuring the number of periods until two pendulums of equal length, but different initial angles, sync up again (i.e. one obtains a phase shift). However, the inaccuracy in this method is very large, in large part due to the amplitude drastically decreasing for large angles as an effect of energy dissipation. Even if a value of $A \approx 0.07$ is obtained, this might therefore not get the "correct value" mark. However, if the amplitude decrease is taken into account, for example by averaging the angular amplitudes over the measurement time, it is possible to obtain a more accurate and correct result.

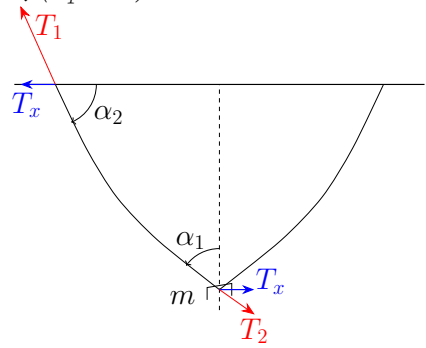
Another common attempt of using a second pendulum as a "clock" results in a far too low time resolution, and therefore is not rewarded any points.

Grading: (preliminary)

Note: The "sync up method" can be rewarded up to 1.5pts (0.5pts for measurements, 1pts for multiple datapoints).

- Recognize that we may use two pendulums with different lengths and angular amplitudes *with equal periods* **(0.5 pts)**
- Presented measurement data of relevant quantities **(0.5 pts)**
- Good choice of measurements (i.e. large enough angles, correct measured angle, long enough pendulum for reasonably large period etc.) **(0.5 pts)**
- Using two or more datapoints (e.g. different lengths etc.) **(1 pts)**
- Obtaining a correct expression for A in terms of measurable quantities **(1 pts)**
- Correct value $A \in [0.06, 0.08]$ **(1.5 pts)**

ii) *(5 points)*



We shall tie some red string to two fixture points on the same horizontal level and dangle the known mass staple from the middle point. We measure the angles the string makes with the horizontal plane at the middle points and next to the supports, averaging the latter for a more accurate value. Let α_1 be the angle at the staple's level, and α_2 be the angle at the support's level. Let T_x be the (unknown) horizontal tension in the string. Along the string, between the staple and the support, T_x is constant, as all acting forces are vertical.

Equilibrium around the staple implies $2T_x \cot \alpha_1 = mg$. Equilibrium on the string strictly between the staple and the support implies $2T_x(\tan \alpha_2 - \cot \alpha_1) = Mg$, where M is the mass of the string. We compute the

[1]Weighing the string on a scale, we found the true val

mass of one meter's worth of string as

$$\begin{aligned}\frac{M}{l} &= \frac{m}{l}\frac{\tan\alpha_2 - \cot\alpha_1}{\cot\alpha_1} \\ &= \frac{m}{l}\left(\tan\alpha_1\tan\alpha_2 - 1\right),\end{aligned}$$

where l is the measured length of string.

We repeat the experiment with different lengths and finally find the average of the results. With the example values $\alpha_1 = 70°$, $\alpha_2 = 41°$, we find $M \approx 0.12$ g.[1]

Grading: (preliminary) Easy to by chance get close to the correct answer using other methods, e.g. scales, balancing, other pendulums, ... Also the method of physical vs mathematical pendulum does not work, likely due to air resistance.

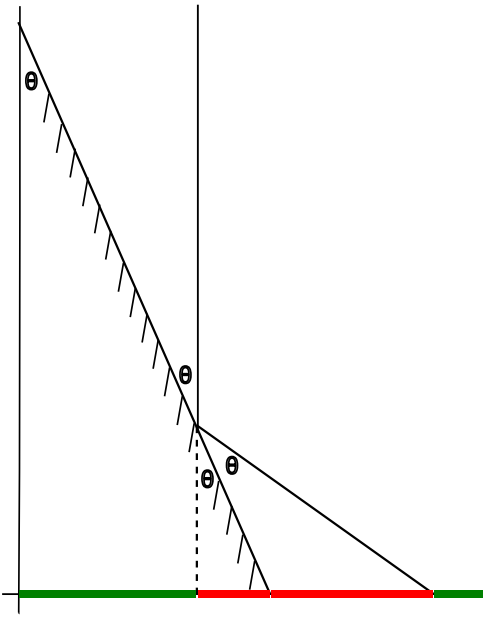
- Recognize the method (banana) **(1 pts)**
- Recognize that the system is in equilibrium, and we may balance the forces **(1 pts)**
- Correct equation **(1 pts)**
- (Conditional on correct method being used) Correct answer [0.11, 0.15] **(2 pts)**

6. Cones (8 points) — *Solution by Tudor Plopeanu, grading schemes by Eppu Leinonen and Aleksi Kononen.*

i) *(2 points)* We will consider any extreme point on the red heart and its reflection. This may be easiest and most precise for the points furthest from the center of the picture, but any edge point of the red heart would suffice. Let θ be the cone's half apex angle. Let r be the measured ratio of the (angular) distance from the perceived image to the mirror and the perceived image to the original point. Then, $\frac{\tan\theta}{\tan 2\theta} = r$, so $\theta = \arccos\frac{1}{\sqrt{2(1-r)}}$, or alternatively $\arctan\sqrt{1-2r}$. Of course, the apex angle is 2θ, which is double the above.

For measured $r = 0.25$, we have the answer $2\theta \approx 1.22$ radians, or 71 degrees.

ue to be 0.135 g.



Grading: (preliminary)

- Stating that the image is formed by vertical rays as the camera is far away **(0.5 pts)**
- Correct explicit expression of θ with respect to r (or some other measurable variable) **(1 pts)** (if not found, partial points can be earned as below)
  - Correct geometrical figure **(0.2 pts)**
  - $r = \tan\theta / \tan 2\theta$ or equivalent or a correct implicit equation for θ **(0.5 pts)**
- Correct numerical answer $2\theta \in [65°, 76°]$ **(0.5 pts)** (only if approach is correct)
  - If only θ is given and not 2θ **(-0.2 pts)**

ii) *(1 point)* The contribution of gravity along the side of the cone has to be equal to the contribution of the centrifugal force. As such, $mg\cos\theta = \frac{mv^2}{R}\sin\theta$, so $v = \sqrt{Rg\cot\theta}$.

Grading: (preliminary)

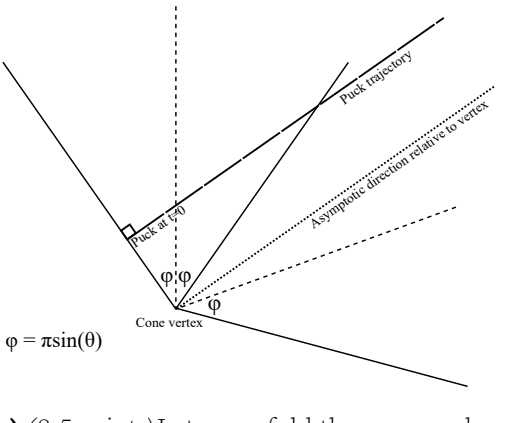
- Correct force balance **(0.5 pts)**
- Correct expression for v **(0.5 pts)**
- Mistakes in trigonometry **(-0.1 pts)**

iii) *(2.5 points)* The puck's energy is the sum of its gravitational potential energy and its kinetic energy, so for a maximal vertical displacement h_f, and the speed v_f at

that moment, we have $E = \frac{v^2}{8} + gh_f = \frac{v_f^2}{2}$. From the conservation of angular momentum between the initial moment and the minimal distance to axis moment, when the velocity is horizontal, we can deduce that $\frac{v}{2}R = v_f R_f$, where R_f is the minimal distance to the cone's axis. We can relate h_f and R_f through $\frac{R-R_f}{h_f} = \tan\theta$. Then, $h_f = (R - R_f)\cot\theta$. Replacing h_f and v_f in the energy conservation, $\frac{v^2}{8} = \frac{v^2R^2}{8R_f^2} - g(R - R_f)\cot\theta$. Replacing $v^2 = Rg\cot\theta$ and solving the third degree polynomial in R_f, $8R_f^3 - 9RR_f^2 + R^3 = 0$, yields the roots R, $\frac{1+\sqrt{33}}{16}R$, and some negative value. Our minimal radius is, as such, $\frac{1+\sqrt{33}}{16}R \approx 0.42R$.

Grading: (preliminary)

- Idea of using energy and angular momentum conservation **(0.2 pts)** (0.1 pts for each)
- Correct energy conservation **(0.5 pts)**
- Correct angular momentum conservation **(0.5 pts)**
- Correct relation between h_f and R_f **(0.1 pts)**
- Correct third degree polynomial **(0.5 pts)**
- Picking the physically relevant solution **(0.5 pts)**
- Correct final answer **(0.2 pts)**



iv) *(2.5 points)* Let us unfold the cone and repeat its pattern, as in the figure above. We note that the relation between the apex angle and the angle of the unfolded cone are re-

lated through the area of a cone $\pi L^2 \sin\theta = \varphi L^2$ and as such $\varphi = \pi \sin\theta$. The puck's trajectory is a straight line starting perpendicular to the edge of our unfolded cone. The radius vector will rotate, in our drawing, by $90° = \pi/2$ over an infinite period of time. To match the rotation on our representation of the cone with the rotation of the radius vector, we remind ourselves that a $360 = 2\pi$ degree rotation of the radius vector matches a 2φ rotation on our drawing. As such, the number of radians of rotation will be $\frac{\pi}{2}\frac{2\pi}{2\pi\sin\theta} = \frac{\pi}{2\sin\theta}$.

Grading: (preliminary)

- Idea of folding out the cone **(0.2 pts)**
- Stating that the trajectory is a straight line in the folded plane **(1 pts)**
- Correct 90 degree rotation in the folded picture or a correct figure **(0.6 pts)**
- Correct relation between rotation in the folded plane and the rotation of the radial vector **(0.7 pts)**
  - Using θ instead of 2θ as the apex angle **(-0.3 pts)**

Solution 2 by Eppu Leinonen

Let ϕ be the total rotation angle of the radius vector and r its length. The speed related to $\dot{\phi}$ is $r\dot{\phi}$ and the speed related to $\dot{r}$ is $\dot{r}\csc\phi$, the projection of which to the radial axis is $\dot{r}$. Thus energy conservation states that $r^2\dot{\phi}^2 + \dot{r}\csc\theta = v^2$. Also the projection of the angular momentum onto the symmetry axis is also conserved and as such $r^2\dot{\phi} = Rv$. From this we solve $\dot{\phi}$ and substitute to the energy conservation equation to get $\dot{r} = v\sqrt{1-(R/r)^2}\sin\theta$. Integrating both sides from time $t = 0$ to $t = t$ ($r(0) = R$ and $r(t) = r$) yields $r^2 = \sqrt{R^2 + v^2t^2\sin^2\theta}$. Then $\dot{\phi} = \frac{Rv}{R^2+v^2t^2\sin^2\theta}$ which we can integrate from $t = 0$ to $t = \infty$ by noticing that $\arctan' x = 1/(1+x^2)$ which gives $\phi_\infty = \frac{\pi}{2\sin\theta}$.

Grading: (preliminary)

- Idea of using energy and angular momentum (projection) **(0.1 pts)** (both required for points)
- Correct energy conservation with respect

to ϕ and r velocity components (can be named freely e.g. v_r and v_ϕ) **(0.4 pts)**
- Correct angular momentum (projection) conservation **(0.4 pts)**
- Equivalent form $\dot{r} = v\sqrt{1-(R/r)^2}$ **(0.5 pts)**
- Solving $r(t)$ (partials below) **(0.5 pts)**
  - Correct integration boundaries **(0.1 pts)**
- Solving ϕ_∞ (partials below) **(0.6 pts)**
  - Setting up integral of the form $\phi_\infty = \int Rv/r^2(t)\mathrm{d}t$ **(0.1 pts)**
  - Correct integration boundaries **(0.1 pts)**
  - $1/\sin\theta$ dependence **(0.3 pts)**

Solution 3 by Eppu Leinonen

Split the velocity into two components: the velocity from changing angle of the radius vector ϕ, $v_\perp$, and the velocity up along the cone $v_\parallel$. In cylindrical coordinates the radial force law now states $N\cos\theta = mv_\perp\dot{\phi} - m\dot{v}_\parallel\sin\theta$ and the vertical force law now states $N\sin\theta = mv_\parallel\cos\theta$. From this we can solve for $\dot{\phi} = \dot{v}_\parallel/(v_\perp\sin\theta)$. From conservation of energy $v_\perp = \sqrt{v^2 - v_\parallel}$ and thus $\phi_\infty = \frac{1}{\sin\theta}\int_0^v \frac{\mathrm{d}v_\parallel}{\sqrt{v^2-v_\parallel^2}} = \frac{\pi}{2\sin\theta}$. The integral can be calculated easily by recoginising that it represents a quarter of the area of a unit circle.

Grading: (preliminary)

- Correct radial and vertical equations of motion **(0.4 pts)** (0.2 for each)
- Representing them in $\phi, v_\parallel, v_\perp$ **(0.4 pts)** (0.2 for each)
- Correct relation between $\dot{\phi}$ and $v_\perp$, $\dot{v}_\parallel$ **(0.5 pts)**
  - For wrong trigonometric dependence **(-0.2 pts)**
- Using the energy conservation law to relate $\dot{\phi}$ to only $v_\perp$ or $v_\parallel$ and derivatives **(0.7 pts)**
- Solving the integral **(0.5 pts)** (partial points below)
  - Correct integration boundaries **(0.1 pts)**
  - Noticing the area of a circle **(0.2 pts)**

Solution 4 by Aleksi Kononen and Eppu Leinonen

Split the velocity into two components: the velocity from changing angle of the radius vector ϕ, $v_\perp$, and the velocity up along the cone $v_\parallel$. Go to the frame corotating with

puck. In this frame, there will be three inertial forces on the puck: centrifugal, Coriolis, and Euler. Both the Coriolis and Euler forces point perpendicularly to $v_\parallel$ and tangentially to the cone. However, as the frame is corotating $\dot{v}'_\perp = 0$ in the frame, and thus the Coriolis and Euler forces must cancel. This means that the radial component of the normal force is $N_r = mv_\perp^2/r$. Thus back in the inertial frame we get $\dot{v}_\parallel = \frac{v_\perp^2}{r}\sin\theta$. On the other hand $\dot{\phi} = v_\perp/r = \frac{1}{\sin\theta}\dot{v}_\parallel/v_\perp$. From energy conservation $v_\perp = \sqrt{v^2 - v_\parallel^2}$ and thus $\phi_\infty = \frac{1}{\sin\theta}\int_0^v \frac{\mathrm{d}v_\parallel}{\sqrt{v^2-v_\parallel^2}} = \frac{\pi}{2\sin\theta}$. The integral can be calculated easily by recoginising that it represents a quarter of the area of a unit circle.

Grading: (preliminary)

- Argument for why the radial component of the normal force is equal $mv_\perp^2/r$ **(0.7 pts)** (partials below for non-inertial frame method)
  - Noting that in the inertial frame there are three inertial forces **(0.1 pts)**
  - Noting that Coriolis and Euler forces are parallel to $v_\perp$ **(0.3 pts)**
  - Noting that Coriolis and Euler forces must be equal as $\dot{v}'_\perp = 0$ **(0.3 pts)**
- Noting that the radial component normal force is equal to $mv_\perp^2/r$ **(0.1 pts)**
- Correct relation between $\dot{\phi}$ and $v_\perp$, $\dot{v}_\parallel$ **(0.5 pts)**
  - For wrong trigonometric dependence **(-0.2 pts)**
- Using the energy conservation law to relate $\dot{\phi}$ to only $v_\perp$ or $v_\parallel$ and derivatives **(0.7 pts)**
- Solving the integral **(0.5 pts)** (partial points below)
  - Correct integration boundaries **(0.1 pts)**
  - Noticing the area of a circle **(0.2 pts)**

7. Waves (4 points) — *Solution by Tudor Plopeanu.*

i) *(1 point)* Both gk^α and $\frac{\sigma}{\rho}k^\beta$'s dimensionalities are s^{-2}, while k's dimensionality is m^{-1}. The surface tension's dimensionality is kg/s^2. As such, $\alpha = 1$ and $\beta = 3$.

Grading: (preliminary)

1. Dimensional analysis (or similar technique). **(0.2 pts)**
2. Units of σ **(0.2 pts)**
3. $\alpha = 1$. **(0.3 pts)**